$$Tr\ |\psi\rangle\langle\psi|\ \mathbf{L}\ = \sum_i \langle i|\psi\rangle\langle\psi|\mathbf{L}|i\rangle.$$

The two factors in the summation are just numbers, so we can reverse their ordering,

$$Tr\ |\psi\rangle\langle\psi|\ \mathbf{L} = \sum_i \langle\psi|\mathbf{L}|i\rangle\langle i|\psi\rangle.$$

Carrying out the sum and using $\Sigma\ |i><i| = I$, we get

$$Tr\ |\psi\rangle\langle\psi|\ \mathbf{L} = \langle\psi|\mathbf{L}|\psi\rangle.$$

The right side is just the expectation value of **L**.

7.3 DENSITY MATRICES: A NEW TOOL

Up to now, we have learned how to make predictions about a system when we know the system's exact quantum state. But more often than not, we don't have complete knowledge of the state. For example, suppose Alice has prepared a spin using an apparatus oriented along some axis. She gives the spin to Bob but doesn't tell him the axis along which the apparatus was oriented. Perhaps she gives him some partial information, such as the fact that the axis was either along the *z* axis or the *x* axis, but she refuses to tell him more than that. What does Bob do? How does he use this information to make predictions?

Bob reasons as follows: If Alice prepared the spin in the state $|\psi>$, then the expectation value of any observable **L** is

$$Tr\ |\psi\rangle\langle\psi|\mathbf{L} = \langle\psi|\mathbf{L}|\psi\rangle.$$

On the other hand, if Alice prepared the spin in state $|\varphi>$, then the expectation value of **L** is